

CVIS

Connected Vehicle Intelligence System

Master Project Documentation

Course: Computer Communication & Network Security (CCNS)
Type: Academic Project — Industry-Inspired Prototype
Version: 2.3 (Phase 5 Ready)
Stack: ESP32 · FastAPI · SQLite · Ollama · Next.js
Protocols: HTTP REST + MQTT (runtime-switchable)
Security: HMAC-SHA256 · AES-256-GCM · API Key Auth

Architecture at a Glance

```
graph LR
    ESP32 -- "HTTP / MQTT" --> FastAPI
    FastAPI -- "HMAC-SHA256 | AES-256-GCM" --> ESP32
    FastAPI -- "WebSocket" --> NextJS[Next.js UI]
    NextJS -- "/driver /noc /admin" --> FastAPI
    NextJS -- "Ollama AI <- Multi-factor Telemetry" --> Ollama[Ollama AI]
```

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Chapter 1

Project Overview

1.1 Project Identity

The **Connected Vehicle Intelligence System (CVIS)** is an academic prototype built for the *Computer Communication & Network Security (CCNS)* course. Its central thesis is that AI reasoning belongs in the cloud, not embedded in each vehicle. The vehicle is a *thin, securely-networked client*; the laptop running the backend acts as the data-centre. CVIS demonstrates the **secure communication layer** that makes this architecture viable.

Core Thesis

AI lives centrally (cloud/data-centre). The vehicle is a thin client. The project grades the **secure communication pipeline**, not the AI itself.

1.2 Core Objectives

1. Simulate a realistic EV telemetry pipeline from edge device to centralised server.
2. Enforce cryptographic integrity (HMAC-SHA256) and optional confidentiality (AES-256-GCM) on every packet.
3. Demonstrate runtime protocol switching between HTTP REST and MQTT publish-subscribe without restarting any service.
4. Provide a live Network Operations Center (NOC) dashboard that visualises packet flow, security events, and chaos simulations in real time.
5. Integrate Ollama AI reasoning over multi-factor telemetry to produce contextually accurate driver recommendations.

1.3 Non-Goals

- Real EV hardware or real sensors — telemetry is simulated on an ESP32 or via the Python physics engine.
- A generic chatbot with no telemetry grounding.
- Three separate frontend codebases (one unified Next.js app serves all three roles).

Chapter 2

System Architecture

2.1 High-Level Architecture Diagram

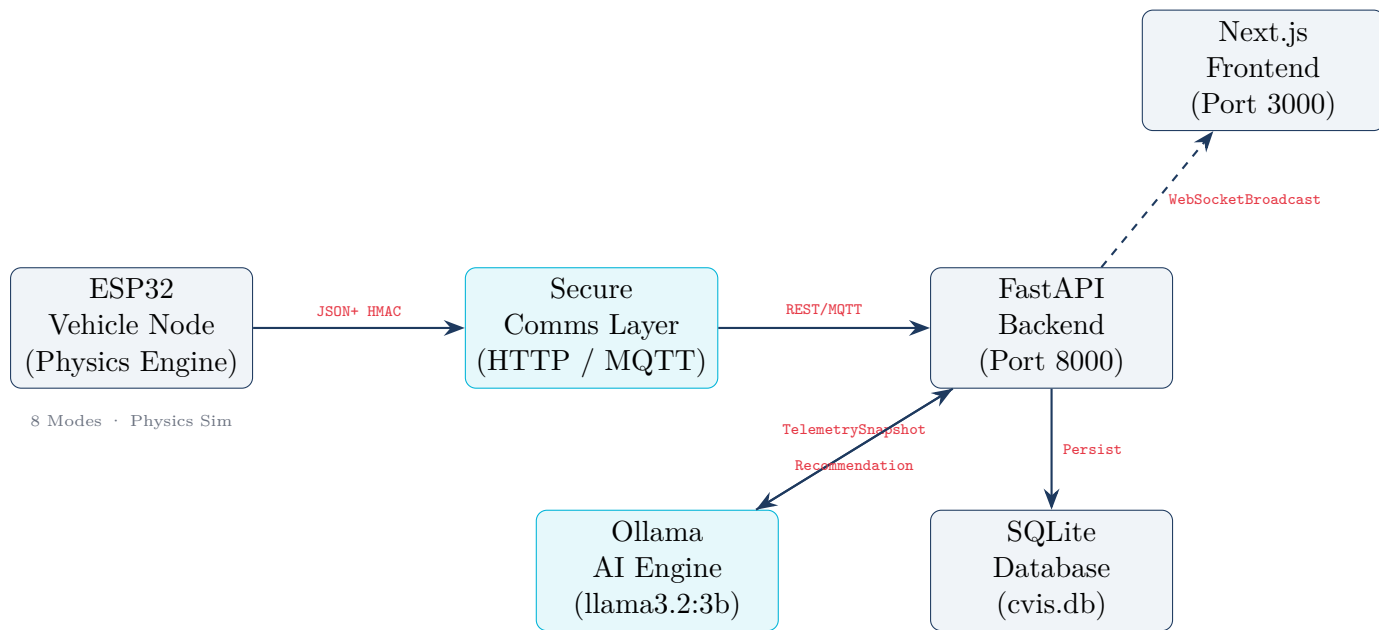


Figure 2.1: CVIS High-Level System Architecture

2.2 Component Summary

Table 2.1: CVIS Component Inventory

Component	Technology	Location	Role
ESP32 Firmware	Arduino C++	<code>/firmware</code>	Stateful physics engine; generates telemetry; signs with HMAC; supports HTTP + MQTT
Python Simulator	Python 3	<code>/backend/simulator.py</code>	Local vehicle physics emulator; same logic as firmware, used for development
FastAPI Backend	Python + FastAPI	<code>/backend</code>	Core server: auth, crypto, chaos middleware, AI bridge, WebSocket fan-out
SQLite Database	aiosqlite	<code>/backend/cvis_db</code>	Persists packets, telemetry, devices, auth logs, and server config
Ollama AI	llama3.2:3b / phi3-mini	Local Ollama	Multi-factor telemetry reasoning, driver recommendations, free-text chat
Next.js Frontend	Next.js + Tailwind	<code>/frontend</code>	Unified three-role UI: Driver, NOC, Admin

Chapter 3

Communication Layer

3.1 Protocol Overview

CVIS supports two fully-functional transport protocols, switchable at runtime with zero restarts:

Table 3.1: Transport Protocol Comparison

Property	HTTP REST	MQTT Pub/Sub
Pattern	Request-Response	Publish-Subscribe
Connection	Stateless, per-request	Persistent TCP via broker
Topic/Endpoint	POST /api/v1/telemetry	cvis/telemetry/{device_id}
Auth mechanism	X-API-Key header	_meta.api_key in envelope
HMAC transport	X-HMAC-Signature header	_meta.hmac in envelope
Config subscribe	Poll /api/v1/config/protocol	Subscribe cvis/config/#
Switch latency	Immediate (next packet)	MQTT broker must be running
Best used when	Demonstration, reliability	Bandwidth-constrained, IoT

3.2 Packet Flow — HTTP

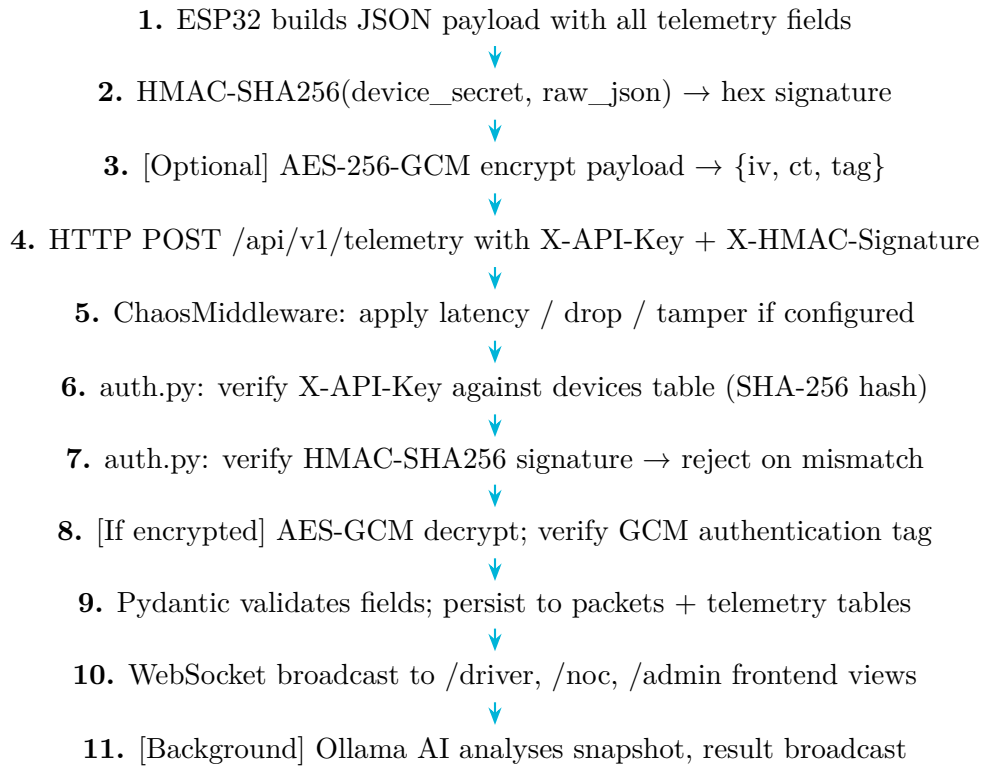


Figure 3.1: HTTP Telemetry Ingest Pipeline

3.3 Packet Schema

Every packet conforms to the following JSON schema (defined in `backend/models.py`):

Table 3.2: Telemetry Payload Schema v1.0

Field	Type	Required	Description & Bounds
device_id	string	Yes	Unique vehicle ID (1–64 chars)
schema_version	string	No	Schema version, default "1.0"
timestamp_ms	integer	Yes	ESP32 <code>millis()</code> at packet creation
mode	string	Yes	One of 8 allowed modes (see Table 5.1)
speed_kmh	float	No	Vehicle speed; $0 \leq v \leq 250$ km/h
battery_pct	float	No	State of charge; $0 \leq \text{SoC} \leq 100\%$
battery_temp_c	float	No	Battery pack temp; $-20 \leq T \leq 100^\circ\text{C}$
motor_temp_c	float	No	Motor temp; $-20 \leq T \leq 150^\circ\text{C}$
range_km	float	No	Estimated remaining range; $0 \leq r \leq 1000$ km
fault_code	integer	No	0x00 = none, 0x02 = overheat, 0x04 = motor
charging_rate_w	float	No	Charging power; 0 when not charging
<i>Advanced Physics Fields (Optional — Schema v2.3)</i>			
ambient_temp_c	float	No	Outside air temperature in $^\circ\text{C}$
headwind_kmh	float	No	Headwind speed in km/h
road_gradient_pct	float	No	Road gradient (+uphill / downhill)
tire_pressure_psi	float	No	Average tire pressure in PSI
cabin_climate_w	float	No	AC/Heater power draw in Watts
max_cell_voltage_diff_mv	float	No	Max voltage difference between cells (V)

Chapter 4

Security Implementation

4.1 Security Layer Architecture

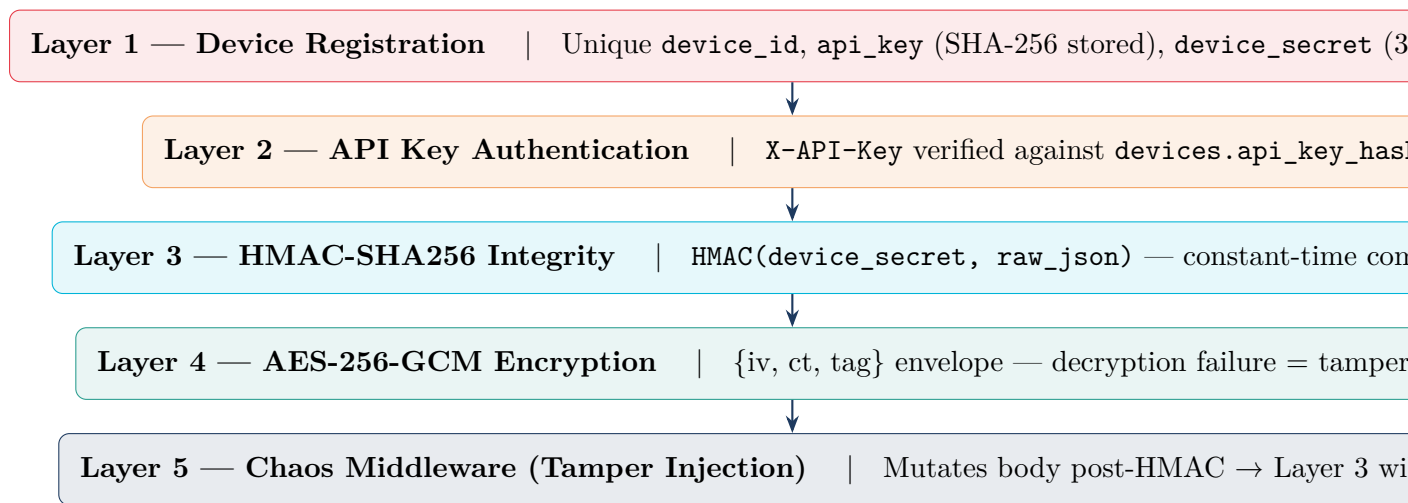


Figure 4.1: CVIS Security Layer Stack

4.2 Device Registration Flow

When the Python simulator or an ESP32 first connects:

1. POST /api/v1/devices/register with {"device_id": "SIM-001"}.
2. Backend generates: api_key = secrets.token_hex(32) and device_secret = secrets.token_hex(32).
3. api_key_hash = SHA256(api_key) is stored in devices table. The plaintext api_key is returned once.
4. All subsequent requests include X-API-Key: <plaintext> and X-HMAC-Signature: <hex>.

4.3 HMAC-SHA256 Implementation

HMAC is computed over the *exact raw JSON bytes* as transmitted. This is critical: even a single extra space in the JSON would change the HMAC and trigger rejection.

Listing 4.1: HMAC Computation (backend/crypto.py)

```

1 def compute_hmac(secret_hex: str, payload: str) -> str:
2     key = bytes.fromhex(secret_hex)          # 32-byte (256-bit) key
3     sig = hmac.new(key, payload.encode("utf-8"), hashlib.sha256)
4     return sig.hexdigest()                  # 64-char lowercase hex
5
6 def verify_hmac(secret_hex: str, payload: str, signature_hex: str) ->
    bool:
7     expected = compute_hmac(secret_hex, payload)
8     # hmac.compare_digest prevents timing-oracle attacks
9     return hmac.compare_digest(expected, signature_hex.strip().lower
        ())

```

4.4 AES-256-GCM Encryption

Listing 4.2: AES-GCM Encrypt/Decrypt (backend/crypto.py)

```

1 def aes_gcm_encrypt(secret_hex: str, plaintext: str) -> dict:
2     key = bytes.fromhex(secret_hex)          # 32 bytes = 256-bit AES key
3     aesgcm = AESGCM(key)
4     nonce = os.urandom(12)                  # 96-bit random nonce per
        message
5     ct_with_tag = aesgcm.encrypt(nonce, plaintext.encode("utf-8"),
        None)
6     ct = ct_with_tag[:-16]                  # ciphertext bytes
7     tag = ct_with_tag[-16:]                 # 128-bit GCM authentication
        tag
8     return {"iv": nonce.hex(), "ct": ct.hex(), "tag": tag.hex()}

```

The GCM **authentication tag** doubles as an integrity check: decrypting with a mismatched key or tampered ciphertext raises `InvalidTag`, which the backend logs as a `tamper_detected` event.

4.5 Fault Codes

Table 4.1: Vehicle Fault Codes

Code (hex)	Name	Description & Physics Trigger
0x00	No Fault	Normal operation
0x02	Battery Overheating	Battery temp > 60°C — caused by high discharge rate under load
0x04	Motor Fault	Motor temp > 95°C — caused by high speed + uphill gradient + drag

Chapter 5

Vehicle Simulation Engine

5.1 Physics Model Overview

The simulator is a **stateful physics engine**. Each tick (every 2 seconds) computes the next state based on the current state. No values are hardcoded per mode; modes are *emergent* from the physics.

5.1.1 Load Factor Computation

The **Load Factor** (L) is the normalised measure of how hard the vehicle is working at any instant:

$$L = \underbrace{\frac{v}{50}}_{\text{speed drag}} + \underbrace{\frac{(v+w)^2}{10000}}_{\text{aerodynamic drag}} + \underbrace{g \cdot 0.5}_{\text{gravity drag}} + \underbrace{\max\left(0, \frac{36 - p_t}{10}\right)}_{\text{tire drag}} \quad (5.1)$$

where v = speed (km/h), w = headwind (km/h), g = road gradient (%), p_t = tire pressure (PSI).

5.1.2 Battery Drain Model

$$\Delta \text{SoC} = \left(L \cdot 0.1 + \frac{P_{\text{cabin}}}{5000} \right) \cdot \Delta t \quad (5.2)$$

5.1.3 Thermal Model

Motor Temperature:

$$\Delta T_{\text{motor}} = (L \cdot 2.0 - (T_{\text{motor}} - T_{\text{ambient}}) \cdot 0.05) \cdot \Delta t \quad (5.3)$$

Battery Temperature:

$$\Delta T_{\text{battery}} = (\Delta \text{SoC} \cdot 5.0 - (T_{\text{battery}} - T_{\text{ambient}}) \cdot 0.02) \cdot \Delta t \quad (5.4)$$

5.1.4 Range Estimation

$$r = \text{SoC} \times 3.0 \times f_{\text{tire}} \times f_{\text{wind}} \times f_{\text{cabin}} \quad (5.5)$$

where penalty factors are: $f_{\text{tire}} = 0.9$ if $p_t < 32$ PSI; $f_{\text{wind}} = 0.85$ if $w > 20$ km/h; $f_{\text{cabin}} = 0.95$ if $P_{\text{cabin}} > 1000$ W.

5.2 Vehicle Operating Modes

Table 5.1: Vehicle Operating Modes — Emergent Classification

Mode	Classification Trigger	Typical Characteristics
Healthy	$30 \leq v \leq 80$ km/h	Normal cruise; moderate load
Eco	(button-selected / low speed)	Conservative speed, max range
Sport	$v > 80$ km/h	High motor temp, fast drain
Heavy Traffic	$0 < v < 30$ km/h	Stop-start, cabin AC dominant
Low Battery	$\text{SoC} \leq 15\%$	Speed limited; urgent AI alert
Battery Overheating	$T_{\text{batt}} > 60^\circ\text{C}$	Fault 0x02; limp mode (40 km/h cap)
Charging	$v = 0$ and $P_{\text{charge}} > 0$	Battery replenishing
Motor Fault	$T_{\text{motor}} > 95^\circ\text{C}$	Fault 0x04; forced stop

5.3 Physics Interdependency Map

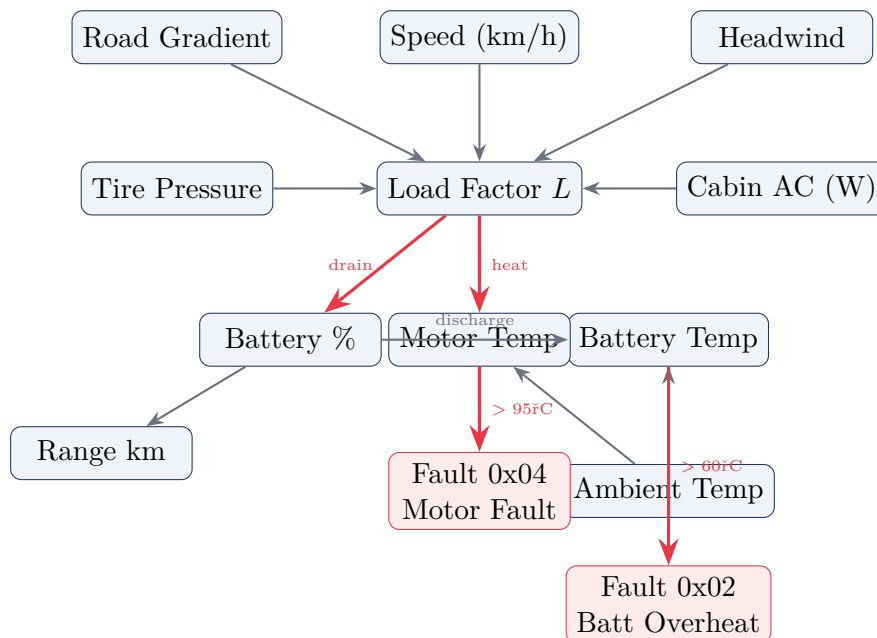


Figure 5.1: Physics Variable Interdependency Graph

Chapter 6

AI Integration

6.1 Architecture

CVIS uses **Ollama** running locally with `llama3.2:3b` (or `phi3-mini`). The AI is called via HTTP to `http://localhost:11434/api/chat`. The model is never called directly from the frontend — it is always mediated by the FastAPI backend, which ensures telemetry grounding.

6.2 Two AI Surfaces

Table 6.1: AI Endpoints

Surface	Endpoint	Behaviour
Auto Recommendation	Background task per telemetry update	Analyses combined snapshot; produces severity-tagged alert; broadcast via Web-Socket
Driver Chat	POST <code>/api/v1/ai/chat</code>	Free-text Q&A grounded in current + last 10 readings; cannot invent vehicle state

6.3 Prompt Engineering

The prompt is built in `backend/ai/prompt_builder.py`. The user message sent to Ollama contains:

1. **Current Snapshot:** all 17 telemetry fields from the latest packet.
2. **Advanced Physics Section:** ambient temp, headwind, road gradient, tire pressure, cabin load, cell voltage delta — labelled `=== ADVANCED PHYSICS ===`.
3. **Trend Block:** last 5 readings rendered as a sparkline text table (battery %, speed, motor temp, mode).
4. **Range Violations:** any field that has crossed its warning threshold.
5. **Mode Transition:** previous mode is explicitly shown when a mode change is detected.

The system prompt instructs the model to: (1) classify severity as **NORMAL/CAUTION/WARNING/CRITICAL**, (2) reason from the *combination* of factors rather than any single threshold, and (3) output exactly two lines: severity label + recommendation.

6.4 AI Throttling

To prevent LLM overload, the background AI task fires only when:

- At least 20 seconds have elapsed since the last inference, **OR**
- The vehicle mode has changed since the last packet.

Chapter 7

Database Schema

7.1 Schema Overview

All data is persisted in a single SQLite file (`backend/cvis.db`) using `aiosqlite` for non-blocking async access. The schema uses WAL (Write-Ahead Logging) for concurrent read-write.

Table 7.1: Database Tables

Table	Primary Key	Purpose
packets	id (auto)	Raw log of every inbound packet; protocol-agnostic
telemetry	id (auto)	Parsed, structured telemetry fields per packet
devices	device_id	Registered vehicle nodes with API key hash and secret
auth_logs	id (auto)	Every auth and tamper event
server_config	key	Key-value store for all runtime configuration
schema_version	version	Migration history tracker

7.2 Packets Table

Listing 7.1: Packets Table DDL

```
1 CREATE TABLE packets (  
2     id                INTEGER PRIMARY KEY AUTOINCREMENT,  
3     received_at       TEXT     NOT NULL,  
4     device_id         TEXT     NOT NULL,  
5     protocol          TEXT     NOT NULL DEFAULT 'http', -- 'http' | '  
6     direction         TEXT     NOT NULL DEFAULT 'inbound',  
7     size_bytes        INTEGER,  
8     status            TEXT     NOT NULL, -- 'ok' | 'tamper_detected'  
9     raw_json          TEXT     NOT NULL,  
10    encrypted         INTEGER NOT NULL DEFAULT 0, -- 0 | 1  
11    encryption_method TEXT     NOT NULL DEFAULT 'PLAIN', -- 'PLAIN' |  
12    'AES-GCM'  
13    auth_status       TEXT     NOT NULL DEFAULT 'ok'  
14 );
```

7.3 Server Config Keys

Table 7.2: Runtime Server Configuration Keys

Key	Default	Description
active_protocol	http	Current transport protocol (http/mqtt)
encryption_enabled	false	AES-256-GCM toggle
auth_enabled	true	API key + HMAC enforcement toggle
replay_protection	false	Timestamp-window replay protection
chaos_loss_pct	0	Packet loss rate (0/5/10/25%)
chaos_latency_ms	0	Artificial latency (0/100/300/1000 ms)
chaos_tamper	false	Payload mutation injector
mobile_app_enabled	true	Global mobile access gate
force_mode	""	Override vehicle simulation mode

Chapter 8

Network Operations Center (NOC)

8.1 NOC Purpose

The NOC at `/noc` is the centrepiece of the CVIS demonstration. It visualises every network event in real time and provides live controls that each trigger a *verifiable backend behaviour change*. No toggle is cosmetic.

8.2 NOC Terminology Glossary

Table 8.1: NOC Terminology Reference

Term	Definition
Packet	A single JSON payload from one vehicle tick, transported via HTTP POST or MQTT PUBLISH
RTT (Round-Trip Time)	Time from packet dispatch to acknowledgement receipt — artificially elevated by the Latency slider
HMAC	Hash-based Message Authentication Code — proves the payload was not modified in transit
AES-GCM	Advanced Encryption Standard in Galois/Counter Mode — provides both confidentiality and integrity
Auth Status	<code>auth_ok</code> <code>auth_fail</code> <code>no_auth</code> <code>tamper_detected</code>
Encryption Status	PLAIN (unencrypted) or AES-GCM (encrypted envelope)
Packet Loss %	Percentage of packets randomly dropped by the ChaosMiddleware before processing
Chaos Latency	Synthetic delay injected per packet in the middleware
Tamper Injection	Mutates a payload field <i>after</i> HMAC is computed, guaranteeing the backend rejects it
Protocol Switch	Live toggle between HTTP REST and MQTT; the backend broadcasts the change via MQTT so ESP32 nodes self-switch
Replay Attack	Re-sending an old captured packet to trick the backend — countered by timestamp-window protection
Limp Mode	Battery Overheating fault state: speed capped at 40 km/h by physics engine
Fan-out	One WebSocket broadcast from backend reaches <i>all</i> connected frontend tabs simultaneously

8.3 NOC Live Controls

Table 8.2: NOC Live Controls — Backend Effect Mapping

Control	API Endpoint	Real Backend Effect
Protocol Switch	POST <code>/api/v1/config/protocol</code>	Changes <code>active_protocol</code> in DB; broadcasts to MQTT <code>cvis/config/protocol</code> ; ESP32 self-switches on next poll
Encryption Toggle	POST <code>/api/v1/config/encryption</code>	Toggles AES-GCM; broadcasts to MQTT; ESP32 switches payload format on next packet
Auth Toggle	POST <code>/api/v1/config/auth</code>	Enables/disables API key + HMAC verification; traffic accepted without credentials in OFF mode
Packet Loss Slider	POST <code>/api/v1/config/chaos</code>	ChaosMiddleware probabilistically returns HTTP 503 before any processing
Latency Slider	POST <code>/api/v1/config/chaos</code>	<code>asyncio.sleep(latency_ms)</code> added before handler runs
Tamper Injector	POST <code>/api/v1/config/chaos</code>	Mutates <code>battery_pct=999.9</code> post-signature; next HMAC check will fail
Replay Protection	POST <code>/api/v1/config/replay</code>	Enables timestamp-window deduplication; duplicate packets rejected
Disconnect Vehicle	POST <code>/api/v1/control/disconnect</code>	Marks device inactive in DB; future packets from that device rejected
Stop AI Service	POST <code>/api/v1/control/ai</code>	Disables background AI task; recommendations paused globally

8.4 NOC Packet Inspector Fields

Each row in the NOC packet table can be expanded to reveal:

- **Packet ID** — sequential integer from `packets.id`
- **Source / Destination** — device ID → backend IP
- **Timestamp** — ISO 8601, backend receipt time
- **Protocol** — `http` or `mqtt`
- **Size (bytes)** — raw JSON body length
- **Auth Status** — `auth_ok` / `auth_fail` / `no_auth` / `tamper_detected`
- **Encryption** — `PLAIN` or `AES-GCM`
- **Fault Code** — hex fault from telemetry
- **AI Response** — last recommendation generated for this device
- **Raw JSON Payload** — full telemetry JSON (collapsible)

Chapter 9

Frontend Pages

9.1 Routing Architecture

The unified Next.js application (port 3000) provides:

Table 9.1: Frontend Routes

URL	Role	Primary Purpose
/	Landing	Role selector, project overview
/driver	Driver	Live vehicle telemetry, AI chat, health gauges, historical graphs
/noc	NOC Engineer	Live packet flow, control panel, security events
/admin	Administrator	System stats, auth logs, device registry, server health
/alphamobile	Premium Driver	Mobile view for ESP32-ALPHA (Premium tier)
/betamobile	Premium Driver	Mobile view for ESP32-BETA (Premium tier)
/gammamobile	Free Driver	Mobile view for ESP32-GAMMA (Free tier — gated)
/deltamobile	Free Driver	Mobile view for ESP32-DELTA (Free tier — gated)

9.2 Driver Dashboard (/driver)

Key Features

- Live telemetry gauges: speed, battery SoC, battery temp, motor temp, range, fault code
- Operating mode badge with colour coding
- AI Recommendation panel (severity-tagged: NORMAL / CAUTION / WARNING / CRITICAL)
- Historical trend graphs (Recharts library): battery%, motor temp, speed over time
- Driver Chat: free-text conversation grounded in live telemetry
- Active alerts panel driven by WebSocket events

Data flow: WebSocket `ws://localhost:8000/ws` → `event: telemetry` message → React state update → re-render gauges.

9.3 NOC Dashboard (/noc)

Key Features

- Live animated packet flow diagram (ESP32 → Backend, colour-coded by protocol and security status)
- Packet table with 10 columns: ID, source, dest, timestamp, protocol, size, auth, encryption, fault, RTT
- Click-to-expand packet inspector (full JSON payload + AI response)
- All 9 live controls wired to real backend endpoints (Table 8.2)
- Live chaos metrics: total requests, dropped packets, tampered packets

9.4 Admin Console (/admin)

Key Features

- Connected vehicles list with status, tier, and last seen
- Server status: uptime, WebSocket client count, AI service health
- Packet statistics chart (last 12 hours by protocol)
- Auth log table: event type, device, IP, timestamp, details
- Failed request tracker and error log
- CPU/memory usage indicators

9.5 Mobile Views (/alphamobile, /betamobile, etc.)

Premium-tier vehicles (ESP32-ALPHA, ESP32-BETA) have a mobile-optimised view accessible at `/alphamobile` and `/betamobile`. Free-tier vehicles (GAMMA, DELTA) see a *gated* page. Access is controlled by the NOC's **Global Mobile App** toggle (`server_config.mobile_app_enabled`), which is verified server-side per device tier before granting access.

Chapter 10

Chaos & Simulation Middleware

10.1 Purpose

The `ChaosMiddleware` (in `backend/middleware/chaos.py`) simulates adverse network conditions at the **application middleware layer** rather than at the OS/network level. This is pedagogically equivalent and far simpler to set up in a demo environment.

10.2 Middleware Pipeline

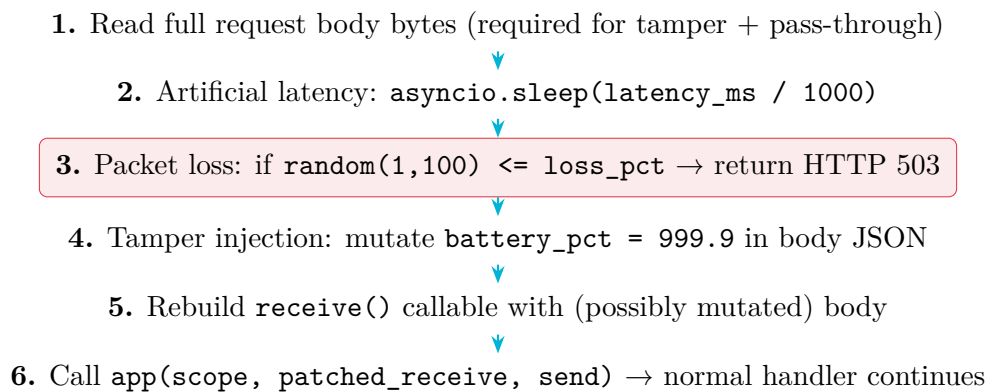


Figure 10.1: ChaosMiddleware Execution Pipeline

10.3 Tamper Detection Sequence

When **tamper injection** is enabled and **auth is enabled**:

1. ESP32 sends JSON with HMAC over the *original* payload.
2. ChaosMiddleware changes `battery_pct`: 999.9 in the body.
3. `auth.py` reads the (now mutated) body and recomputes HMAC.
4. HMAC mismatch detected → `tamper_detected` event logged to `auth_logs`.
5. Backend returns HTTP 401 with "HMAC signature verification failed".
6. NOC packet table shows `auth_status = tamper_detected` in red.

Chapter 11

API Reference

11.1 Complete Endpoint Inventory

Table 11.1: Full API Endpoint Reference

Endpoint Description	Method
Telemetry	
/api/v1/telemetry Ingest telemetry packet (HTTP); requires X-API-Key + X-HMAC-Signature	POST
/api/v1/telemetry/recent Fetch N most recent telemetry rows; optional device_id filter	GET
/api/v1/telemetry/packets Fetch raw packet log for NOC table	GET
Devices	
/api/v1/devices/register Register new device; returns api_key + device_secret	POST
/api/v1/admin/deactivate-device Mark device as inactive	POST
Configuration (NOC Controls)	
/api/v1/config/protocol Get/set active protocol (http/mqtt)	GET/POST
/api/v1/config/encryption Get/toggle AES-GCM encryption	GET/POST
/api/v1/config/auth Get/toggle API key + HMAC enforcement	GET/POST
/api/v1/config/chaos	GET/POST

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Endpoint Description	Method
Get/update packet loss, latency, tamper settings <code>/api/v1/config/replay</code>	GET/POST
Get/toggle timestamp replay protection <code>/api/v1/config/all</code>	GET
Full config snapshot (all settings in one call) <code>/api/v1/config/mode</code>	POST
Force a specific vehicle simulation mode <code>/api/v1/config/mobile-access</code>	GET
Get mobile access state for a vehicle <code>/api/v1/config/mobile-app</code>	POST
Globally enable/disable mobile app access	
AI	
<code>/api/v1/ai/recommendation</code> On-demand AI recommendation for latest telemetry	POST
<code>/api/v1/ai/chat</code> Driver chat grounded in current + historical telemetry	POST
<code>/api/v1/ai/status</code> Check Ollama availability and model status	GET
Admin	
<code>/api/v1/admin/stats</code> System stats: uptime, packet counts, auth counts	GET
<code>/api/v1/admin/devices</code> Connected vehicle list with status and tier	GET
<code>/api/v1/admin/packet-stats</code> Packet statistics over last N hours	GET
<code>/api/v1/admin/auth-logs</code>	GET

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Endpoint Description	Method
Authentication and tamper event log	
Control	
/api/v1/control/vehicles List all vehicles with real-time status	GET
/api/v1/control/disconnect Disconnect / deactivate a vehicle	POST
/api/v1/control/ai Enable/disable AI service globally	POST
WebSocket & System	
/ws WebSocket endpoint; fan-out broadcast to all connected clients	WS
/health Backend health check	GET
/docs Swagger UI auto-generated API docs	GET

Chapter 12

CCNS Syllabus Mapping

Table 12.1: CVIS Features Mapped to CCNS Units

Unit Where in Code	Topic	CVIS Feature
Unit 1 main.py, ws_manager.py	Network Topologies, Client-Server	ESP32 (thin client) → FastAPI (server); NOC, Admin, Driver roles
Unit 2 comms/mqtt_adapter.py, routers/config.py	Application Layer Protocols, REST, MQTT Pub/Sub	HTTP REST + MQTT runtime-switchable; MQTT topics cvis/telemetry/# and cvis/config/#
Unit 3 middleware/chaos.py, firmware.ino	Reliability, RTT, Packet Loss, Latency, Retransmission	ChaosMiddleware: configurable loss, latency, tamper; ESP32 retry with exponential backoff
Unit 4 auth.py, crypto.py, routers/devices.py	Authentication, Integrity, Encryption	API Key Auth, HMAC-SHA256, AES-256-GCM, tamper detection logging

Unit Where in Code	Topic	CVIS Feature
Unit 4 replay_protection.py, routers/config.py	Replay Attack Protection	Timestamp-window deduplication; same (device_id, timestamp_ms) rejected within window
Unit 5 ai/ollama_client.py, ai/prompt_builder.py	Cloud/Edge AI, Intelligent Networking	Ollama AI centralised on server; multi-factor reasoning; not embedded per-vehicle
Unit 5 ws_manager.py, routers/ws.py	Real-time Data Streaming	WebSocket fan-out; single broadcast reaches all three frontend views simultaneously

Chapter 13

Startup & Deployment

13.1 Prerequisites

Table 13.1: Software Prerequisites

Software	Version	Purpose
Python	≥ 3.11	Backend and simulator runtime
Node.js	≥ 18	Next.js frontend
Ollama	Latest	Local AI model runtime
Mosquitto MQTT	≥ 2.0	Optional MQTT broker (needed for MQTT mode)
Arduino IDE / PlatformIO	Latest	For flashing ESP32 firmware

13.2 One-Click Startup

A platform-specific launcher script starts all services and opens browser tabs automatically:

- **macOS:** `start_cvis.command` (double-click in Finder)
- **Windows:** `start_cvis.bat` (double-click in Explorer)

Both scripts: (1) activate Python venv, (2) start FastAPI backend, (3) start Next.js frontend, (4) start Python physics simulator, (5) open four browser tabs: `/driver`, `/noc`, `/admin`, `/alphamobile`.

13.3 Port Map

Table 13.2: Service Port Map

Port	Service	Notes
8000	FastAPI Backend	REST + WebSocket
3000	Next.js Frontend	All three UI roles
11434	Ollama AI	HTTP API (<code>/api/chat</code>)
1883	Mosquitto MQTT	Optional

Chapter 14

Security Analysis

14.1 Threat Model

Table 14.1: Threat Model and Mitigations

Threat	CCNS Category	CVIS Mitigation
Packet tampering in transit	Integrity	HMAC-SHA256: any bit change fails verification; logged as <code>tamper_detected</code>
Payload eavesdropping	Confidentiality	AES-256-GCM encryption; ciphertext is opaque without 32-byte device secret
Impersonation / spoofing	Authentication	Per-device API key; SHA-256 stored (never plaintext)
Replay attack	Non-repudiation	Timestamp-window replay protection; duplicate (<code>device_id</code> , <code>ts</code>) rejected
Rogue device	Access Control	Device must register first; inactive devices are rejected
AI hallucination	Data Integrity	Prompt is strictly grounded in real telemetry; model cannot invent state

14.2 Cryptographic Strength

- **API Keys:** 256-bit random (`secrets.token_hex(32)`), stored as SHA-256 hash.
- **HMAC Keys** (`device_secret`): 256-bit random per device.
- **AES-GCM Nonces:** 96-bit random per message (the GCM standard); never reused.
- **Timing Attacks:** `hmac.compare_digest` used for constant-time comparison.

Appendix A

WebSocket Event Types

Table A.1: WebSocket Event Schema

Event Key	Payload Description
telemetry	Full telemetry snapshot for all 17 fields; broadcast on every successful ingest
ai_recommendation	{device_id, severity, recommendation} — broadcast after AI inference
packet_logged	{packet_id, device_id, protocol, auth_status, encrypted, size_bytes}
global_mobile_app_changed	{changed: bool} — broadcast when NOC changes mobile access gate

Appendix B

Fault Code Reference

Table B.1: Extended Fault Code Reference

Code	Name	Physics Trigger	AI Response Expectation
0x00	No Fault	—	NORMAL severity unless other thresholds breached
0x02	Battery Overheating	$T_{\text{batt}} > 60^{\circ}\text{C}$	WARNING or CRITICAL; AI explains load factors causing heat
0x04	Motor Fault	$T_{\text{motor}} > 95^{\circ}\text{C}$	CRITICAL; AI recommends immediate stop; references speed + gradient

Appendix C

Physics Engine — Key Constants

Table C.1: Physics Engine Constants

Constant	Value	Use
Base drain coefficient	0.1	Battery% per unit load per second
Cabin drain coefficient	1/5000	Battery% per Watt per second
Motor heating coefficient	2.0	°C per unit load per second
Motor cooling coefficient	0.05	°C decay towards ambient per second
Battery heating coefficient	5.0	°C per unit drain per second
Battery cooling coefficient	0.02	°C decay towards ambient per second
Range base multiplier	3.0	km per 1% battery at ideal conditions
Motor fault threshold	95°C	Above → Fault 0x04 injected
Battery overheat threshold	60°C	Above → Fault 0x02 injected
Low battery threshold	15% SoC	Below → Mode = "Low Battery"
Limp mode speed cap	40 km/h	Applied when Fault 0x02 is active